Resource Pack — Datasets, Papers & Tools

Satellite-Based Early Warning for Glacial Lake Outburst Floods (GLOF) · Major Project — 7th Semester

Everything the team needs, in one place. All datasets are public and free; total data cost is zero.

> **Do this in Week 1:** the main GLOFNet dataset is hosted on an informal Google Drive folder, not a permanent archive. Download it and keep two independent backup copies before starting any other work. If that folder disappears, the project loses its foundation.

1. PRIMARY DATASET

Name	What it is / why we need it	Link
GLOFNet (2025)	The core multimodal GLOF-monitoring dataset: 600+ Sentinel-2 patches (128×128 px, 6 bands at 10 m), ~11.9 million ITS_LIVE glacier velocity records (1988–2024), and ~98,000 MODIS land-surface-temperature observations for Shisper Glacier, Karakoram, 2000–2024. Its authors list our exact objectives as their own future work.	Paper: arxiv.org/abs/2510.10546 Data: Google Drive folder linked from the paper

2. SATELLITE DATA SOURCES

Source	What it gives us	Link / Earth Engine ID
Sentinel-2 L2A (optical)	10 m colour imagery — used to measure lake area, shoreline and surface change. Revisits every ~5 days but is blocked by cloud.	dataspace.copernicus.eu GEE: COPERNICUS/S2_SR_HARMONIZED
Sentinel-1 GRD (SAR radar)	All-weather, day-and-night radar that sees through monsoon cloud — the key addition of this project.	GEE: COPERNICUS/S1_GRD search.asf.alaska.edu
ITS_LIVE (NASA MEaSUREs)	Global glacier surface velocity from the 1980s to present — gives the "glacier is speeding up" signal.	its-live.jpl.nasa.gov
MODIS MOD11A1 v6.1	Daily 1 km land surface temperature — drives the melt-season thermal index.	lpdaac.usgs.gov/products/mod11a1v061 GEE: MODIS/061/MOD11A1
Copernicus DEM 30 m / NASADEM	Elevation model for slope, aspect and moraine geometry features.	GEE: COPERNICUS/DEM/GLO30 GEE: NASA/NASADEM_HGT/001

3. LAKE INVENTORIES AND FLOOD RECORDS (LABELS)

Source	What it gives us	Link
HMA GLOF Database	697 documented glacial lake outburst floods across High Mountain Asia, 1833–2022, with dates, locations, lake type, discharge and impacts. India accounts for ~8.5% of events. This is our label source for the retrospective test.	doi.org/10.5281/zenodo.7271187
ICIMOD glacial lake inventories	Regional inventories across the Hindu Kush Himalaya, including which lakes are classed as potentially dangerous — used to pick our 15–25 study lakes.	lib.icimod.org/records/q27cs-w1b80
GLIMS / Randolph Glacier Inventory	Glacier outlines, used to link each lake to its parent glacier.	glims.org/RGI

4. PAPERS AND ARTICLES TO READ

#	Title & why it matters	Link
1	**GLOFNet: A Multimodal Dataset for Glacial Lake Outburst Flood Monitoring** (2025) — our primary dataset paper; read the "limitations and future work" section closely, it defines our novelty.	arxiv.org/abs/2510.10546
2	**Sequence of events that led to the South Lhonak lake outburst flood in Sikkim, India** — *Scientific Reports*. The step-by-step account of the disaster we are testing against.	nature.com/articles/s41598-026-35895-7
3		doi.org/10.1126/science.ads2659

	The Sikkim flood of October 2023: drivers, causes and impacts of a multihazard cascade — *Science*. The authoritative analysis of the trigger.	
4	**The likely causes of the 4 October 2023 Sikkim flood disaster** — The Landslide Blog, *Eos*. Readable plain-English explanation, good for the introduction and the viva.	eos.org/thelandslideblog/sikkim-flood-1
5	**A comprehensive and version-controlled database of GLOFs in High Mountain Asia** — *ESSD* 15, 3941–3961 (2023). The paper behind the HMA GLOF database.	essd.copernicus.org/articles/15/3941/2023
6	**Comprehensive analysis of the catastrophic 4 October 2023 GLOF at South Lhonak Lake** — ScienceDirect. Detailed forensic reconstruction.	sciencedirect.com/science/article/pii/S2214581825004033
7	**Glacial lakes and GLOFs in a warming Himalaya–Karakoram region** — *npj Natural Hazards*. Best single overview of the field; use it for the literature survey.	nature.com/articles/s44304-026-00168-w
8	**The Status of Glacial Lakes in the Hindu Kush Himalaya** — ICIMOD Research Report 2018/1. Regional context and the "potentially dangerous lake" criteria.	lib.icimod.org/records/q27cs-w1b80

5. TOOLS, PLATFORMS AND DOCUMENTATION

Tool	Used for	Link
Google Earth Engine	Server-side satellite filtering, compositing and export — this is what keeps the project inside free compute limits.	earthengine.google.com (free research tier)
Google Colab	Model training and experiments on a free T4 GPU.	colab.research.google.com
Microsoft Planetary Computer	STAC catalogue for direct scene access without downloading whole archives.	planetarycomputer.microsoft.com
rasterio / rioxarray / GDAL	Reading, writing and reprojecting satellite rasters.	rasterio.readthedocs.io
xarray + Dask	Handling the multi-dimensional data cube without running out of memory.	docs.xarray.dev
geopandas / shapely	Lake polygons, basin boundaries, spatial joins.	geopandas.org
scikit-learn	Isolation Forest, One-Class SVM, PCA — the anomaly-detection baselines.	scikit-learn.org/stable/modules/outlier_detection.html
PyTorch	Temporal autoencoder / LSTM forecasting models.	pytorch.org
Streamlit + Folium + Plotly	The dashboard: risk map, time-series explorer, alert configuration.	streamlit.io · folium docs
MLflow	Logging experiments and keeping a model registry.	mlflow.org
GitHub (Student Pack)	Single shared repository — code, notes and drafts in one place.	education.github.com/pack

6. WHERE TO PUBLISH

Venue type	Examples	Why
Workshops (try first)	NeurIPS / ICML / ICLR Climate Change AI workshops; EGU and AGU sessions	~40–60% acceptance, fast review, real feedback
Journals (later extension)	Remote Sensing of Environment; Natural Hazards and Earth System Sciences (NHESS); IEEE JSTARS	Suitable once the benchmark and results are complete
National	INCOIS / ISRO symposia; Indian disaster-management conferences	Strong regional relevance and direct access to the user community

Note: dataset availability, hosting and licences should be re-verified by the team at the start of the project. Links were correct as of the proposal date.